

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
MATHEMATICS DEPARTMENT
Mathematics for Natural Science (math1011) (Test I (15%))

Part I (Short answer) Write your simplified answer on the space provided (1pts for each blank space)

- The negation of $(\forall x)(3x+1 < 0)$ is $(\exists x)(3x+1 \geq 0)$
- All real number x and some real number y such that $x^2 - y^2 > 0$. The open statement symbolically is $(\forall x)(\exists y)(x^2 - y^2 > 0)$ and the truth value is F
- Let $A = \{2x : -3 < x \leq 3, x \in \mathbb{Z}\}$ and $B = \{x : 0 < x < 6, x \in \mathbb{Z}\}$, then
 - List $p(A \cap B) = \{2\}$
 - $n(A/B) = \{-2, 0\}$
- Let $\text{LCM}(a, 45) = 180$ and $\text{GCF}(a, 45) = 3$, then the value of $a = 12$
- Let $z = \frac{2}{\sqrt{3}} - 2i$ and $w = -\frac{2}{\sqrt{3}} + 2i$, then
 - The principal argument of zw is $\frac{\pi}{3}$
 - The polar form of z^4 is $2^{\frac{16}{9}} (\cos(\frac{4\pi}{3}) - i \sin(\frac{4\pi}{3}))$
 - Solve $w^{1/2}$. Ans: $\frac{1}{\sqrt{3}} + \sqrt{3}i, -\frac{1}{\sqrt{3}} - i\sqrt{3}$

Part II (Workout): Show all the necessary steps clearly and neatly. (2+2+2 pts)

- Identify the premises, write the argument form and check the validity of the argument:
If he studies medicine, he will get a good job. If he gets a good job, he will get a good wage. He did not get a good wage. Therefore, he did not study medicine.

p : he studies medicine
 q : he will get a good job
 r : he got a good wage

Arg: $p \Rightarrow q, q \Rightarrow r, \neg r \vdash \neg p$

- $\neg r \equiv T$ premise
 $r \equiv F$

- $q \Rightarrow r \equiv T$ premise
 $q \Rightarrow F \equiv T$
 $q \equiv F$

- $p \Rightarrow q \equiv T$ premise
 $p \Rightarrow F \equiv T$
 $p \equiv F$

- $\neg p \equiv T$

$\therefore p \Rightarrow q, q \Rightarrow r, \neg r \vdash \neg p$ is valid

@keleme-2013

2. Using mathematical induction prove that $\sum_{i=1}^n i^3 = \frac{1}{4}n^2(n+1)^2$; $\forall n \in \mathbb{N}$.

$$n=1 \quad 1^3 = \frac{1}{4}(1)(4) = 1$$

$$n=k \quad 1^3 + 2^3 + \dots + k^3 = \frac{1}{4}(k)^2(k+1)^2$$

$$n=k+1 \quad 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

$$\frac{1}{4}k^2(k+1)^2 + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

$$\frac{1}{4}(k^2 + 2k^2 + k^2) + k^3 + 3k^2 + 3k + 1 = \frac{1}{4}(k^2 + 2k + 1)(k^2 + 4k + 4)$$

$$\frac{1}{4}k^4 + \frac{6k^3}{4} + \frac{13k^2}{4} + 3k + 1 = \frac{k^4 + 6k^3 + 13k^2 + 12k + 4}{4}$$

$$\frac{1}{4}k^4 + \frac{6k^3}{4} + \frac{13k^2}{4} + 3k + 1 = \frac{k^4}{4} + \frac{6k^3}{4} + \frac{13k^2}{4} + 3k + 1$$

$$\frac{k^2}{4}(k^2 + 6k + 13) + 3k + 1 = \frac{k^2}{4}(k^2 + 6k + 13) + 3k + 1$$

3. For all integers a and b, if a is even and b is odd, then 4 not divide $a^2 + b^2$

i) Write the contrapositive of the statement

statement: $P(x) = (p \wedge q) \Rightarrow \neg r$

contrapositive: $\neg(\neg r) \Rightarrow \neg(p \wedge q)$

$r \Rightarrow \neg(p \wedge q)$

ii) Prove the statement by contradiction

$$a = 2n$$

$$b = 2n+1$$

$$a^2 = 4n^2$$

$$b^2 = (2n+1)^2 = 4n^2 + 4n + 1$$

$$a^2 + b^2 = 8n^2 + 4n + 1$$

$$\frac{a^2 + b^2}{4} = 2n^2 + n + \frac{1}{4}$$

Since there is a remaining $\frac{1}{4}$ $a^2 + b^2$ is not divided by 4

p - a is even

q - b is odd

r - 4 divide $a^2 + b^2$